
23CSE202 DATABASE MANAGEMENT SYSTEMS

S3 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

PROJECT ABSTRACT

Group No

C6

Project Title

Online Food Ordering System

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ABSTRACT

Traditional food ordering through phone calls or in-person visits is slow and challenging for restaurants to manage, leading to lost orders and a poor customer experience. The Online Food Ordering System solves this problem by using a database-driven application that automates the process of ordering food from restaurants through a single online platform. With the rapid growth of internet services and digital technologies, customers now prefer ordering food online due to its convenience, time-saving nature, and accessibility from anywhere at any time. This project aims to design and develop a Database Management System (DBMS) that stores, organizes, retrieves, and manages the data required for an online food ordering platform, acting as a bridge between customers, restaurants, delivery partners, and administrators.

The system is intended for four categories of users:

- 1. Customers** – who create accounts, browse through restaurant menus, search for food items, add products to a cart, place orders, make online payments, and track order status.
- 2. Staff** – who add or modify menu items, manage incoming and outgoing orders, and prepare orders for delivery.
- 3. Delivery Partners** – who accept delivery requests, update the delivery status, and deliver orders to customers.
- 4. Administrators** – who view customer details, restaurant information, payment records, delivery partner information, and delivery data across the platform.

By keeping all data in a relational database, it ensures the data is organized, accurate, consistent, secure, and easy to access while avoiding duplicate records.

The database is designed using an Entity-Relationship (ER) model, relational schema, and normalization to organize data efficiently. It uses primary keys, foreign keys, and integrity constraints to maintain accurate relationships between entities such as Customers, Restaurants, Food Items, Orders, Payments, Deliveries, and Administrators. SQL queries and joins are used to retrieve information such as order details and sales reports. Transactions ensure that order placement, payment, and inventory updates are completed correctly. Triggers automatically update food item availability and order status, while stored procedures perform common tasks such as calculating order totals and delivery charges. These features help to maintain data accuracy, consistency, and reliability.

The primary objective of this project is to demonstrate the practical application of Database Management System (DBMS) concepts by solving a real-world business problem. The defined entities, relationships, and core modules make the system easy to design and implement within the given time period. The database is designed to support the main functions of an online food ordering platform in an efficient and organized manner. It helps to process orders quickly and securely store customer, restaurant, payment, and delivery information, improving the accuracy and consistency of data. The system also makes it easier to manage restaurant operations like tracking orders, generating reports, and retrieving order information whenever needed.

In conclusion, the Online Food Ordering System is a practical implementation of Database Management System (DBMS) concepts, demonstrating how a well-designed database can support the daily operations of an online food ordering platform. The project is feasible to design, implement, and test within the given timeline, covering the essential features required for managing customers, restaurants, delivery partners, orders, payments, and administrators. It shows how proper database design improves data organization, security, and overall system performance, highlighting the importance of techniques such as ER modeling, normalization, relational schema design, SQL queries, transactions, and stored procedures in building an efficient database.

This project provides valuable hands-on experience in database systems and implementation while strengthening the understanding of core DBMS concepts. It also demonstrates how database concepts can be used to solve real-world problems and support the development of scalable and efficient applications. Overall, the project serves as a strong foundation for learning database concepts and applying them in future software development projects.